

Zeyad El-Sayed

AI/ML Engineer | LLM & Agentic AI Systems

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Cairo, Egypt | Military Service: Completed (Jul 2024 – Dec 2025)

SUMMARY

AI/ML Engineer specializing in production **LLM and agentic systems**. Built an autonomous coding agent, a research automation pipeline, and a multilingual RAG platform, taking each from design through deployment. Published **4 NLP models** in the official Spark NLP Model Hub (John Snow Labs).

SKILLS

Languages: Python, C++, SQL

LLM & Agentic AI: LangChain, LangGraph, ReAct Agents, Multi-Provider Routing & Failover, Tool Calling, Prompt Engineering, RAG, Hybrid Retrieval, Query Rewriting, Cross-Encoder Reranking, PEFT (LoRA/QLoRA)

ML Frameworks: PyTorch, HuggingFace Transformers, Scikit-learn, Keras, NumPy, Pandas

Backend & APIs: FastAPI, Async APIs, SQLAlchemy, Alembic, Pydantic, JWT Auth, REST, PostgreSQL, Redis

Infra & DevOps: Docker, Docker Compose, GitHub Actions (CI/CD), Linux, uv, Textual (TUI), Railway, Supabase, Cloudflare Workers

Vector & Retrieval: pgvector, FAISS, BM25, Reciprocal Rank Fusion, Semantic Search

PROJECTS

TesseractCLI — Autonomous Coding Agent

Jul 2026 – Present | GitHub

Independent Developer | Python, LangChain-core, Textual, Pydantic, LLM Routing, GitHub Actions

- Built a terminal-based coding agent from scratch, using a custom tool loop and sandboxed execution instead of an existing framework.
- Established failover routing across 11 LLM providers, automatically retrying requests whenever a provider fails.
- Integrated approval gates and automated security scans into every commit to catch risky actions early.

TesseractRAG — Hybrid Retrieval Platform

Mar – Apr 2026 | GitHub | Live Demo

Independent Developer | FastAPI, PostgreSQL, pgvector, BM25, Redis, Docker

- Redesigned the storage layer around PostgreSQL and pgvector for reliable, consistent data handling.
- Combined keyword and vector search into a hybrid pipeline, reranked for accuracy across Arabic and English.
- Introduced user accounts and usage-based routing to keep API costs under control.

TesseractResearch — Agentic Research Platform

Jul 2026 | GitHub | Live Demo

Independent Developer | LangGraph, FastAPI, AsyncPostgreSQL, Redis, Docker

- Architected a 6-stage research agent that validates its own outputs and retries failed steps automatically.
- Streamlined multi-hour research runs with save-and-resume checkpoints, ensuring progress is never lost.
- Deployed an async backend that streams live progress to users, packaged as a 4-service Docker stack.

Catch the AI (Graduation Project)

May 2024 | GitHub | Website

Python, Django, React, PostgreSQL, ViT, DeBERTa, Wav2Vec 2.0

- Trained an image classifier to detect AI-generated images at 96% accuracy.
- Fine-tuned language models to detect AI-generated text at 95% accuracy; three were later published in Spark NLP.
- Extended the system with an audio detection model reaching 90% accuracy, covering all three modalities.

EXPERIENCE

Information Technology Institute (ITI) Business Intelligence Trainee

Jul 2023 – Aug 2023 Egypt

- Developed a SQL data warehouse with dimensional modeling and ETL pipelines to structure raw data.
- Delivered interactive Tableau and Power BI dashboards on top of the warehouse.

EDUCATION

Benha University B.S. in Computer Science & Artificial Intelligence

Oct 2020 – Jun 2024 GPA: 3.4 / 4.0

LANGUAGES

Arabic: Native | **English:** Professional working proficiency